

Software Requirements Specification (SRS)

AI Financial Auditing Platform

1. Introduction

1.1 Purpose

This document describes the software requirements for the AI Financial Auditing Platform, a system that automates financial auditing processes using Artificial Intelligence and Machine Learning.

The platform aims to assist auditors, accountants, and regulatory bodies in detecting financial anomalies, ensuring regulatory compliance, and improving audit efficiency.

1.2 Scope

The system will provide:

- Automated financial data analysis
- AI-based anomaly detection
- Compliance verification (Tax & Accounting standards)
- Risk scoring of financial transactions
- Audit report generation
- Dashboard for auditors and financial analysts

The platform targets:

- Accounting firms
- Financial auditors
- Enterprises
- Government regulators
- Tax authorities

1.3 Definitions, Acronyms, Abbreviations

AI – Artificial Intelligence

ML – Machine Learning

NLP – Natural Language Processing

OCR – Optical Character Recognition

Audit Trail – Record of all system actions

ZATCA – Saudi Tax Authority

VAT – Value Added Tax

1.4 Intended Audience

This document is intended for:

- Software engineers
- System architects
- Project managers
- Financial auditors
- Investors

- Regulatory authorities

2. Overall Description

2.1 Product Perspective

The system is a cloud-based financial auditing platform that integrates:

- Accounting data
- Financial transactions
- Invoice systems
- Tax compliance data
- AI models

The platform processes financial records automatically and produces audit insights and risk indicators.

2.2 Product Functions

Main functions include:

1. Upload financial data
2. AI financial analysis
3. Fraud detection
4. Tax compliance verification
5. Audit reporting
6. Dashboard visualization
7. Risk scoring
8. Document analysis

2.3 User Classes

Auditor:

- Review audit results
- Generate reports
- Verify anomalies

Permissions: Access financial analysis and generate audit reports

Financial Manager:

- Upload financial records
- Review compliance reports

Permissions: View dashboards and upload documents

System Administrator:

- Manage users
- Configure AI models
- Monitor system activity

Permissions: Full system access

2.4 Operating Environment

Frontend: Web Browser, Mobile responsive UI

Backend: Python, Django Framework

Database: PostgreSQL

AI Components: Python, OpenAI models, machine learning libraries

Infrastructure: Cloud servers, Docker containers

2.5 Design Constraints

- Compliance with financial regulations
- Data privacy laws
- High security requirements
- Scalability for large financial datasets

2.6 Assumptions and Dependencies

Assumptions: Users provide valid financial data; integration with accounting systems is available.

Dependencies: AI models, OCR services, cloud infrastructure.

3. System Features

3.1 Financial Data Import

The system shall allow users to upload financial data in formats such as Excel, CSV, PDF, and accounting system exports.

Features: Data validation, Data mapping, Automatic classification.

3.2 AI-Based Financial Analysis

The system shall analyze financial data using AI algorithms to detect abnormal transactions, unusual financial patterns, and accounting inconsistencies.

Outputs include anomaly reports, financial insights, and risk indicators.

3.3 Fraud Detection

The system shall identify potential fraud using pattern recognition, anomaly detection, and behavioral analysis.

Examples include duplicate invoices, suspicious payments, and irregular transactions.

3.4 Compliance Checking

The system shall verify compliance with VAT regulations, accounting standards, and tax requirements.

Examples: incorrect tax calculations, missing invoices, regulatory violations.

3.5 Audit Report Generation

The system shall generate automated audit reports including financial risk summary, anomaly detection results, and compliance status.

Report formats: PDF, Excel, Dashboard view.

3.6 Dashboard

The system shall provide dashboards displaying financial risk score, anomalies detected, compliance status, and audit history.

Visualization: charts, graphs, indicators.

4. External Interface Requirements

4.1 User Interface

The platform will provide web-based dashboards, interactive charts, and a data upload interface.

Frameworks: Django Templates, TailwindCSS, Alpine.js.

4.2 API Interface

The platform will expose APIs for accounting software, financial systems, and external audit tools.

4.3 Hardware Interface

No specialized hardware required. Users access system via computers, tablets, and mobile devices.

5. Non Functional Requirements

5.1 Performance

The system shall process financial datasets up to 1 million records, generate audit reports within 10 seconds, and support multiple concurrent users.

5.2 Security

Security requirements include encrypted data transmission, role-based access control, audit logs, and multi-factor authentication.

5.3 Reliability

The system shall ensure 99.9% uptime, automatic backups, and fault tolerance.

5.4 Scalability

The system must scale to support large enterprises, multi-company environments, and cloud deployment.

5.5 Maintainability

The system architecture should support modular AI components, easy updates, and microservice compatibility.

6. AI Requirements

6.1 Anomaly Detection Models

AI models shall detect anomalies in financial transactions, expense patterns, and revenue patterns.

Techniques: Isolation Forest, Neural Networks, Statistical models.

6.2 Document AI

The system shall analyze documents including invoices, receipts, and financial statements using OCR and NLP.

6.3 Risk Scoring

The AI system shall calculate Financial Risk Score based on transaction anomalies, compliance violations, and historical patterns.

7. Future Enhancements

- Predictive financial risk analysis
- Real-time transaction monitoring
- Blockchain audit trails
- Integration with ERP systems

8. Appendix

Example data sources include accounting systems, bank statements, invoices, and tax records.